MINISTRY OF EDUCATION AND TRAINING
CAN THO UNIVERSITY

INFORMATION ON RESEARCH RESULTS

1. General information:

Project code: THS2026-68

Project title: Development of an Edge AI-based System for Multimodal Analysis and Clustering of Flood Rescue Events

Project coordinator: Lê Thị Ngọc Ảnh

Implementing institution: Trường Công nghệ Thông tin và Truyền thông, Đại học Cần Thơ

Duration: May–October 2026 (six months).

2. Objective(s):

Develop an AI-based mobile system to receive and analyze flood rescue information and support decision-making under unstable network conditions, with the aim of reducing loss of life and property damage.

3. Creativeness and innovativeness:

The project integrates edge image inference, adaptive transmission and event clustering in a mobile–server–map prototype. Location, time and report content support grouping and human verification.

4. Research results:

On 256 test images, the seed-1024 baseline achieved 75.78% accuracy and 74.26% macro-F1 with three critical errors. Historical pruning reached 80.08% accuracy; exact McNemar $p = 0.1173$ did not establish improvement. A 5/10/20% grid selected 5% using validation, but test macro-F1 fell to 68.14%. Benchmarking covers 32 configurations on four phones. CPU corruption screening does not establish field-camera robustness.

The English CrisisMMD pilot reaches text/fusion macro-F1 of 76.60%/78.35%. The Vietnamese translation/AI-label pilot uses 179/58/59 train/dev/test pairs; fusion reaches 64.91% macro-F1 and 33.33% request recall. DistilBERT fine-tuning averages 52.29% macro-F1 across three seeds. The team confirms that the Vietnamese translations and AI

labels have been checked and that their wording is close to native Vietnamese rescue messages. Fusion superiority is not established. The project team confirms that the mobile team has integrated the NLP/fusion branch into the mobile application. These metrics were measured offline.

At 50 kbit/s and 600 ms round-trip time, metadata and compressed images succeeded in 257/257 attempts, versus 230/257 (89.49%) for originals. These were computer-based proxy measurements. The team confirms that the application exit or session-loss issue when submitting reports offline has been fixed and no longer occurs.

RQ1 does not establish Product superiority over independently selected Additive (ARI 0.9072 versus 0.9191; Holm $p = 0.2560$). RQ2 supports exact-copy invariance under fixed grouping without general alignment gains (NDCG@5 0.6669 versus random 0.6763); three raw Spearman and five Holm values remain unreconciled. RQ3 favors nearest-first on simulated harm and deadline misses. Field rescue benefits remain unmeasured.

5. Products:

The project registers three main scientific and technological products:

– A mobile application supporting disaster rescue, integrating AI models and rescue request submission functions.

– A web-based rescue information management system featuring a digital map for visualizing rescue locations and clustering related events.

– A trained and optimized multimodal AI model package for analyzing rescue information and supporting deployment on mobile devices.

In addition, the project registers a research bulletin, a research summary and a demonstration video of no more than two minutes, in accordance with Can Tho University's requirements. The paper "Stress-Testing Product-Gated Clustering and Bounded Priority Ranking for Flood-Rescue Reports: A Synthetic Study" has been published in the ISDS 2026 proceedings, Part II, in Communications in Computer and Information Science (CCIS), volume 3253, Springer Nature, 2026.

6. Effects, technology transfer means and applicability:

Planned transfer supports education and research at Can Tho University. Field applicability requires professional validation; rescue effectiveness remains unmeasured.